

UPSKILL CAMPUS

Free Summer Internship - Data Science

Project Report

Project Title: **Customer Churn Prediction using Machine Learning**

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Internship Period: Summer 2026

Organization: upSkill Campus & Uniconverge Technologies (P) Ltd.

1. Objective

The main objective of this project is to predict which customers are likely to "churn" or leave a company. Using Machine Learning, we analyze customer data to identify patterns and help the company take steps to retain customers.

2. Dataset Description

Dataset used: Customer data with features like Age, Gender, Tenure, Monthly Charges, Contract Type, and Churn status.

Total Rows: ~7000

Total Columns: 21

3. Data Preprocessing & EDA

- *Data Cleaning*: Handled missing values, removed duplicates
- *Data Encoding*: Converted categorical data to numerical using LabelEncoder and OneHotEncoder
- *Feature Scaling*: Used StandardScaler for normalization
- *EDA*: Plotted bar graphs and heatmaps to find correlation between tenure, charges and churn rate

4. Model Building & Results

Algorithms Used: Logistic Regression, Random Forest, Decision Tree

Best Model: Random Forest

Accuracy: 82%

Key Insight: Customers with low tenure and month-to-month contracts have higher churn rate.

5. Conclusion

This project successfully predicted customer churn with 82% accuracy. The model can help businesses identify at-risk customers and reduce churn by offering targeted retention strategies.

6. [GitHub Repository Link](https://github.com/Vaishnavi%20kesarwani/upskillcampus)

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